A SET-LEVEL MAP OF SHEAVES WHOSE STALK MAPS ARE MORPHISMS OF ALGEBRAIC STRUCTURES IS A MORPHISM OF SHEAVES OF ALGEBRAIC STRUCTURES

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Allowed standard prerequisites (editorial).

- The map F(U) -> prod F_x is injective for a sheaf of sets (lemma-sheaf-subset-stalks). invoked at proof lines 1657-1659. Provides the injectivity of the bottom horizontal arrow.
- If F(g) injective and Im(F(f)) subset Im(F(g)) then f factors through g, in the setting of algebraic structures with forgetful functor F (lemma-image-contained-in). invoked at proof lines 1660-1662. Provides the factorization making the left vertical arrow a morphism of algebraic structures.
- The worked application of the image-containment lemma to this square situation (example-application-lemma-image-contained-in). invoked at proof lines 1660-1663. Illustrates the factorization step applied to the present diagram.

Source reference. Original lemma 0079: sources/stacks-sheaves/pinned-original-source.tex, source lines 1162–1162. Cited at proof lines 1657-1659 for the injectivity of the bottom horizontal arrow; load-bearing inside the counted span.

Source reference. Original lemma 007P: sources/stacks-sheaves/pinned-original-source.tex, source lines 1480–1480. Cited at proof lines 1660-1662 to factor the left vertical arrow; load-bearing inside the counted span.

Source reference. Original remark 007Q: sources/stacks-sheaves/pinned-original-source.tex, source lines 1499–1499. Cited at proof lines 1660-1663 alongside the lemma as the worked instance of the factorization argument; supporting reference inside the counted span.

Lemma 0.2. *let X be a topological space. Let $(\mathcal{C}, F)$ be a type of algebraic structure. Suppose that $\mathcal{F}$, $\mathcal{G}$ are sheaves on X with values in $\mathcal{C}$. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a map of the underlying sheaves of sets. If for all points $x \in X$ the map $\mathcal{F}_x \to \mathcal{G}_x$ is a morphism of algebraic structures, then φ is a morphism of sheaves of algebraic structures.*

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Proof. Let U be an open subset of X. Consider the diagram of (underlying) sets

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \prod_{x \in U} \mathcal{F}_x \\ \downarrow & & \downarrow \\ \mathcal{G}(U) & \longrightarrow & \prod_{x \in U} \mathcal{G}_x \end{array}$$

By assumption, and previous results, all but the left vertical arrow are morphisms of algebraic structures. In addition the bottom horizontal arrow is injective, see Lemma 0079. Hence we conclude by Lemma 007P, see also Example 007Q $\square$